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Unsupervised clustering on MBTI personalities

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Abstract

This paper explores the use of unsupervised learning methods for identifying personality-related structure in textual data. Using user-generated forum posts labeled with Myers–Briggs personality types, we focus on distinguishing between the ENFP and INTJ personality categories. Two unsupervised clustering algorithms, K-Medoids and Affinity Propagation, are evaluated using both handcrafted linguistic features and TF–IDF representations. Although ground-truth labels are available, they are used exclusively for evaluation. Experimental results show that the best clustering configuration achieves a purity of approximately 77%, approaching the performance of supervised baselines trained on the same data. These findings suggest that meaningful personality-related patterns can be partially recovered from text using unsupervised methods, despite the inherent ambiguity of the task.

1 Introduction

Personality traits shape how individuals perceive the world and express themselves through language. The Myers–Briggs Type Indicator (MBTI) is a widely used personality framework that categorizes individuals into sixteen types based on four dimensions: *Extraversion–Introversion* (E/I), *Sensing–Intuition* (S/N), *Thinking–Feeling* (T/F), and *Judging–Perceiving* (J/P). Each personality type is associated with characteristic cognitive and communication patterns.

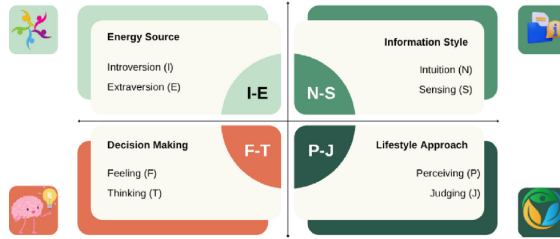


Figure 1: The four MBTI dichotomies (Hartono et al. [3])

This project investigates whether such personality-related structure can be recovered from text using unsupervised machine learning methods. Rather than predicting the four MBTI dimensions independently, we focus on distinguishing between two nearly opposing personality types: ENFP and INTJ. ENFPs are commonly described as expressive and people-oriented, while INTJs tend to be more introspective and structured, suggesting that stylistic and linguistic differences may emerge in writing.

The task is inherently challenging, as textual data captures only a partial view of personality and significant overlap exists between personality types. Moreover, there is no established benchmark for unsupervised clustering of MBTI personalities, since most prior work formulates the task as supervised classification. We apply two unsupervised clustering algorithms K-Medoids and Affinity Propagation to user social media posts text, using both engineered linguistic features and TF–IDF representations. Ground-truth labels are used strictly for evaluation and interpretation.

While supervised approaches achieve high performance, with BERT-based models reaching accuracies of up to 94% [2] and traditional classifiers around 85%–89% [5], our best unsupervised configuration attains a purity of approximately 77%. This performance is comparable to supervised baselines such as Logistic Regression and Support Vector Classification trained on the same data, both achieving around 79% accuracy.

2 Dataset

The experiments are conducted using the Myers–Briggs Personality Type Dataset [6] available on Kaggle. The dataset contains approximately 8,600 user entries, each consisting of a personality type label and a textual field comprising the last 50 posts written by the user on the PersonalityCafe forum.

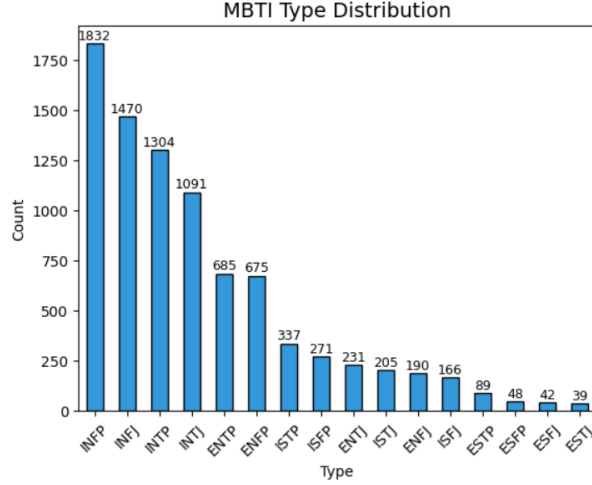


Figure 2: Class distribution of MBTI dataset

The original dataset is highly imbalanced when considering all sixteen MBTI personality types. For the purposes of this study, we restrict the analysis to users labeled as ENFP and INTJ, two personality types that differ across multiple MBTI dimensions. Within this subset, the INTJ class contains a larger number of samples (1,091) compared to the ENFP class (675). To avoid the effects of class imbalance and ensure fair evaluation, the majority class was undersampled, resulting in a balanced dataset with an equal number of samples per class.

3 Preprocessing

The preprocessing of the text samples was minimal, with removing links and extra white space. An important step was also removing MBTI tags to prevent data leakage.

```
1 def preprocess_text(text):
2     # Remove MBTI type mentions
3     text = re.sub(r'\b(intj|infj|enfj|entj|intp|infp|enfp|entp|istj|isfj|
4     estj|esfj|istp|isfp|estp|esfp)\b', '', text, flags=re.IGNORECASE)
5     # Remove URLs
6     text = re.sub(r'https?://\S+|www.\S+', '', text)
7     # Remove extra whitespace
8     text = ' '.join(text.split())
9     return text
10
11 df_balanced['posts_clean'] = df_balanced['posts'].apply(preprocess_text)
```

4 Feature Representations

Two distinct feature representation strategies were employed to capture different aspects of linguistic expression of different personality types. The first approach leveraged handcrafted linguistic features designed to capture stylistic patterns that might differentiate between personality types. These included: punctuation frequency (commas, exclamation marks, question

marks), paragraph structure metrics, emoticon usage, sentence length distributions and word length statistics (all the features are displayed in Table 1).

The second approach utilized TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to capture content-based lexical patterns. The vocabulary was limited to the 1000 most informative unigrams and bigrams based on TF-IDF scores, balancing representational capacity with computational efficiency.

Both feature sets were subsequently subjected to dimensionality reduction to address the curse of dimensionality and improve model generalization. Three techniques were applied:

- **PCA (Principal Component Analysis):** A linear dimensionality reduction method that identifies the principal axes of variation in the data. By projecting the features onto these orthogonal axes, PCA preserves the global structure of the dataset while filtering out noise and redundancy.
- **SVD (Singular Value Decomposition):** A matrix factorization technique that decomposes the feature matrix into three component matrices. This decomposition effectively captures the most significant patterns in the data through singular vectors and values, similar to PCA but without requiring mean-centering of the data.
- **UMAP (Uniform Manifold Approximation and Projection):** A nonlinear dimensionality reduction method that models the data as existing on a lower-dimensional manifold. It assumes there’s a simpler, underlying structure to the features, and it tries to find and represent that simpler structure.

Each reduced representation was evaluated separately to determine which combination of feature extraction and dimensionality reduction yielded the most discriminative signals for personality type classification.

4.1 Handcrafted Linguistic Features

Twenty-six handcrafted linguistic features were extracted from each text sample to capture stylistic patterns potentially indicative of personality differences.

Table 1: Handcrafted Linguistic Features

Feature	Description	Cat.
Sentence length metrics	avg_sentence_length, sentence_rhythm_cv	S
Punctuation frequency	ellipsis, question, parentheses, dash rates	S
Paragraph structure	para_length_cv	S
Exclamation metrics	exclamation_rate, multiple_exclamation_rate	L/E
Visual expression	emoticon_rate, caps_words_rate	L/E
Formal punctuation	colon_rate, semicolon_rate	L/E
Content markers	number_rate, quote_rate	L/E
Expressiveness	expressive_punct, short/long_sentence_rate, comma_rate	E
Vocabulary complexity	avg_word_length, long/very_long_word_rate	V
Lexical diversity	unique_word_ratio	V
Punctuation ratios	excl_to_period, question_to_statement, struct_vs_express	I

Note: Categories: S=Structure, L/E=Logic/Emotion, E=Expressive, V=Vocabulary, I=Interaction

4.2 TF-IDF Representation

The TF-IDF (Term Frequency–Inverse Document Frequency) features were generated with the following preprocessing parameters:

```
1 TfIdfVectorizer(max_features=1000, min_df=5, max_df=0.95,  
2                  stop_words='english')
```

These settings resulted in a sparse feature matrix of shape $1000 \times N$, where N is the number of text samples. The vocabulary was limited to the 1000 most informative unigrams and bigrams, filtering out terms that appeared in fewer than 5 documents or in more than 95% of documents.

5 Unsupervised Models

For personality type clustering, two unsupervised learning algorithms were selected: K-Medoids and Affinity Propagation. These models represent distinct approaches to cluster discovery, each with different assumptions about cluster structure and formation.

- **K-Medoids:** A partition-based algorithm that requires the number of clusters k to be specified in advance. It selects actual data points (medoids) as cluster centers.
- **Affinity Propagation:** An algorithm that automatically determines the number of clusters by exchanging messages between data points. The algorithm identifies exemplars (cluster centers) based on a similarity matrix, with cluster count controlled by a *preference* parameter that influences how many exemplars are selected.

The combination of these approaches allows for comparison between a fixed-cluster model (K-Medoids) and a model that discovers cluster structure autonomously (Affinity Propagation).

5.1 K-Medoids Clustering

K-Medoids, introduced by Kaufman and Rousseeuw [4], is a partition-based clustering algorithm that groups data by selecting representative data points called medoids. Unlike K-Means, which uses calculated centroids, K-Medoids chooses actual observations as cluster centers, providing enhanced robustness to outliers. The algorithm operates by iteratively refining medoid selections to minimize the total dissimilarity between points and their assigned medoids.

Algorithm 1 K-Medoids Clustering Algorithm

```
1: Randomly select  $k$  initial medoids  $M = \{m_1, m_2, \dots, m_k\}$  from  $X$   
2: repeat  
3:   Assignment step: Assign each point  $x \in X$  to nearest medoid  
4:    $C(x) \leftarrow \arg \min_{m \in M} d(x, m)$   
5:   Update step:  
6:   for each medoid  $m \in M$  do  
7:     for each non-medoid  $o \in X \setminus M$  do  
8:       Compute total cost if  $m$  and  $o$  are swapped  
9:       if swap reduces total cost then  
10:        Accept swap:  $M \leftarrow (M \setminus \{m\}) \cup \{o\}$   
11:       end if  
12:     end for  
13:   end for  
14: until No medoid swaps reduce total cost  
15: return  $M, C$ 
```

The total dissimilarity cost for a configuration is defined as:

$$J = \sum_{i=1}^k \sum_{x \in C_i} d(x, m_i)$$

where C_i represents the set of points assigned to medoid m_i , and $d(x, m_i)$ denotes the distance between point x and medoid m_i .

5.2 Affinity Propagation

Affinity Propagation, introduced by Frey and Dueck [1], is an algorithm that identifies exemplars (cluster centers) through message passing between data points.

5.2.1 Algorithm Overview

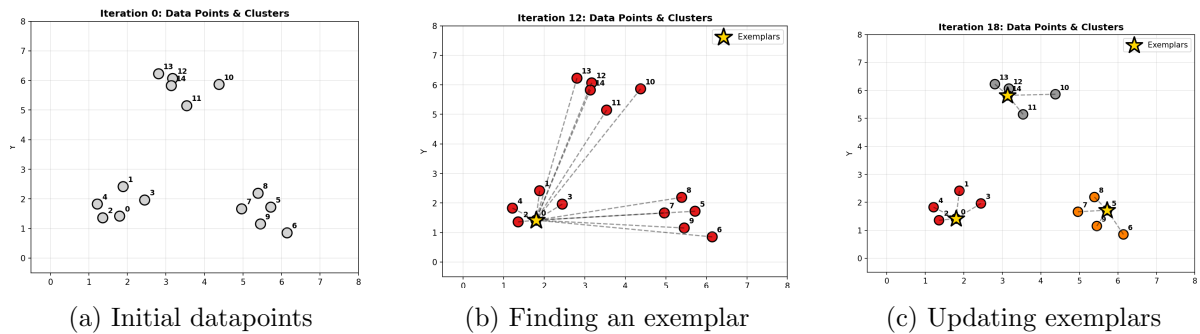
The algorithm operates by using and updating the values from the following matrices:

- **Similarity Matrix S :** $S(i, k)$ represents how well point k serves as an exemplar for point i , typically computed as the negative squared Euclidean distance: $S(i, k) = -\|x_i - x_k\|^2$ for $i \neq k$. The diagonal entries $S(k, k)$ represent each point's preference to serve as an exemplar, with higher values increasing this likelihood.
- **Responsibility Matrix R :** $R(i, k)$ reflects the accumulated evidence that point k should be the exemplar for point i , considering other potential exemplars.
- **Availability Matrix A :** $A(i, k)$ represents the accumulated evidence that point i should choose point k as its exemplar, considering the support from other points.

5.2.2 Message Passing Mechanism

1. **Responsibility Update:** Each point sends *responsibility* messages to candidate exemplars, indicating how suitable each candidate is relative to alternative choices.
2. **Availability Update:** Candidate exemplars send *availability* messages back to points, reflecting how well-suited they are to serve as exemplars considering support from other points.

These updates create a feedback loop: points compete to become exemplars by gathering positive responsibilities, while exemplars consolidate their position by accumulating high availabilities.



5.2.3 Convergence and Exemplar Identification

The algorithm iterates until the messages converge or a maximum number of iterations is reached. Exemplars are identified as points k where:

$$R(k, k) + A(k, k) > 0$$

Each non-exemplar point i is then assigned to the exemplar k that maximizes $A(i, k) + R(i, k)$.

6 Experimental Setup

Features were standardized using `StandardScaler()` to ensure equal contribution during model training. Scalers were fit exclusively on training data and applied to test data to prevent data leakage. To evaluate the impact of dimensionality reduction, multiple distinct configurations were created for both feature types.

Table 2: Experimental Design: Feature Configurations by Type

Configuration	Feature Type		Components
	TF-IDF	Handcrafted	
Raw (no scaling)	✓	✓	Full set
Scaled only	✓	✓	1000 / 26
PCA (2, 5, 10 components)	✓	✓	2, 5, 10
PCA + Scaled	✓	✓	2, 5, 10
TruncatedSVD (2, 5, 10 components)	✓	✓	2, 5, 10
TruncatedSVD + Scaled	✓	✓	2, 5, 10
UMAP (2, 5, 10 components)	✓	✓	2, 5, 10
UMAP + Scaled	✓	✓	2, 5, 10

The configurations enable systematic comparison between:

- Raw versus standardized features
- Reduced features with and without post-reduction scaling
- Different dimensionality reduction techniques (PCA, SVD, UMAP)
- Varying numbers of retained components (2, 5, 10)
- TF-IDF (1000 features) versus handcrafted features (26 features)

All dimensionality reduction methods were initialized with `random_state=42` to ensure reproducible results. Reducers were fit on training data only and used to transform both training and test sets.

6.1 Evaluation Framework

We assessed clustering quality using three complementary metrics combined into a weighted score:

- **Silhouette Score:** Measures cluster cohesion versus separation, ranging from -1 to 1 (higher is better).
- **Cluster Purity:** Percentage of the dominant class within each cluster.
- **Adjusted Rand Index (ARI):** Quantifies similarity between predicted clusters and true labels, adjusted for chance agreement.

6.2 Comparing with External Baselines

To contextualize clustering performance, we compared against:

- **Random baseline:** 3,000 trials of random cluster assignments to establish statistical significance.
- **Supervised baselines:** Logistic Regression and SVM trained on the same feature representations, comparing accuracy (supervised) against purity (unsupervised).

7 Hyperparameter Tuning

Hyperparameter tuning was conducted on the training set using a grid search approach across all feature configurations. The data was split into 80% training and 20% test sets with stratification to maintain class balance.

7.1 Search Space

Table 3 summarizes the hyperparameter search space for each clustering algorithm.

Table 3: Hyperparameter Search Space

Algorithm	Parameter	Values
K-Medoids	Distance metric	euclidean, manhattan, cosine
	Initialization	k-medoids++, heuristic
Affinity Propagation	Damping	0.5, 0.7, 0.9
	Preference	-800, -500, -200, -100, -50

For K-Medoids, the number of clusters was fixed at $k = 2$ to match the binary classification task (INTJ vs. ENFP). For Affinity Propagation, configurations producing fewer than 2 or more than 20 clusters were filtered out to ensure meaningful cluster structures.

7.2 Selection Criterion

Model selection was based on a combined score that balances multiple evaluation metrics:

$$\text{Combined Score} = 0.4 \times \text{Purity} + 0.4 \times \text{ARI} + 0.2 \times \text{Silhouette} \quad (1)$$

This weighting scheme was chosen because:

- **Purity (40%):** Measures how well clusters align with true personality types, directly relevant to our classification goal.
- **ARI (40%):** Penalizes configurations that achieve high purity through excessive cluster counts, as ARI adjusts for chance agreement and rewards configurations where cluster boundaries match true label boundaries.
- **Silhouette (20%):** Ensures clusters are geometrically well-separated, providing a label-independent quality measure.

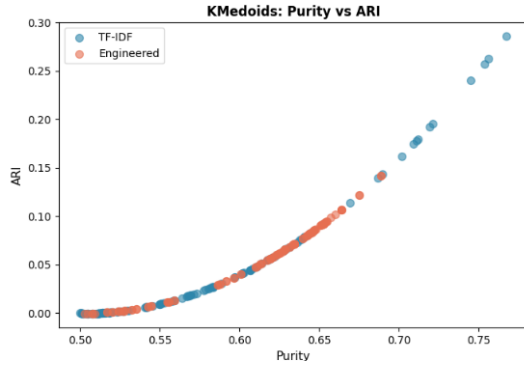
Equal weighting of Purity and ARI specifically addresses Affinity Propagation’s tendency to produce many small, high-purity clusters that do not generalize well.

8 Results

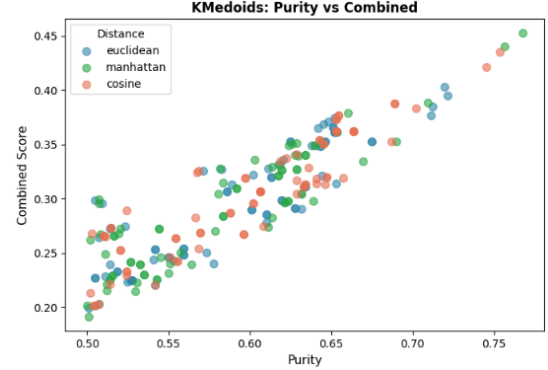
Table 4: Top 10 Configurations from Hyperparameter Tuning (Training Set)

Rank	Configuration	Algorithm	Purity	ARI	Sil.	Combined
1	TFIDF_SVD-5	KMedoids	0.768	0.286	0.155	0.452
2	TFIDF_SVD-5_Scaled	KMedoids	0.756	0.262	0.164	0.440
3	TFIDF_PCA-5	KMedoids	0.754	0.257	0.154	0.435
4	TFIDF_PCA-5_Scaled	KMedoids	0.745	0.240	0.135	0.421
5	TFIDF_SVD-5	KMedoids	0.719	0.192	0.192	0.403
6	TFIDF_SVD-5_Scaled	AffinityProp (k=5)	0.726	0.158	0.239	0.402
7	TFIDF_PCA-5_Scaled	KMedoids	0.721	0.195	0.140	0.395
8	TFIDF_PCA-5_Scaled	AffinityProp (k=2)	0.721	0.195	0.140	0.395
9	TFIDF_SVD-5	KMedoids	0.709	0.174	0.176	0.389
10	ENG_SVD-2	KMedoids	0.689	0.142	0.277	0.388

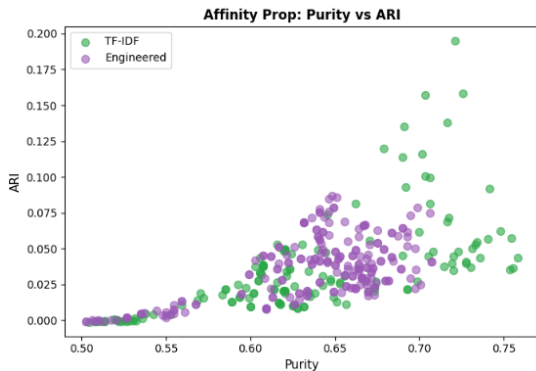
- **TF-IDF dominance:** 9 out of 10 top configurations use TF-IDF features, with only one engineered feature configuration (ENG_SVD-2) appearing in the top 10.
- **Dimensionality reduction:** SVD and PCA with 5 components consistently outperformed other reduction methods, suggesting that 5 dimensions capture sufficient variance for personality type separation.
- **Algorithm preference:** K-Medoids dominated the top results (8 out of 10), with Affinity Propagation appearing twice when producing a small number of clusters (k=2 and k=5).



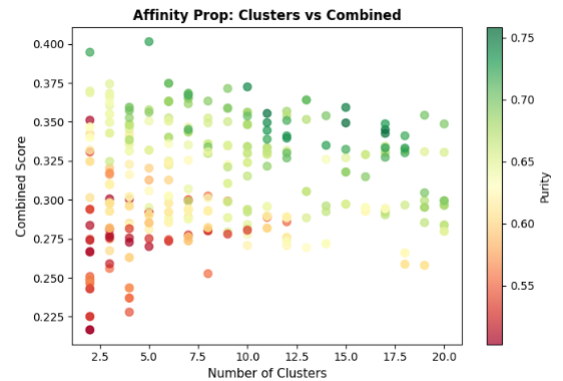
(a) KMedoids: Purity vs ARI



(b) KMedoids: Purity vs Combined



(c) Affinity Prop: Purity vs ARI



(d) Affinity Prop: Purity vs Combined

8.1 Test Set Evaluation

The best configurations from hyperparameter tuning were evaluated on the held-out test set (20% of data). Table 5 presents the results.

- **Generalization:** Both algorithms generalized well to unseen data.
- **KMedoids superiority:** KMedoids with TFIDF_SVD-5 features achieved the best test performance (77.0% purity, 0.290 ARI), outperforming Affinity Propagation across all metrics except silhouette score.
- **Cluster structure:** Affinity Propagation produced 5 clusters on the test set, achieving higher silhouette score (0.229 vs 0.157) but lower ARI (0.175 vs 0.290), indicating better-separated but less label-aligned clusters.

Table 5: Best Configuration Performance on Test Set

Algorithm	Configuration	Parameters	Purity	ARI	Sil.	Combined
KMedoids	TFIDF_SVD-5	manhattan, k-medoids++	0.770	0.290	0.157	0.455
AffinityProp	TFIDF_SVD-5_Scaled	damping=0.7, pref=-500, k=5	0.733	0.175	0.229	0.409

8.2 Comparison with Baselines

To contextualize clustering performance, we compared the unsupervised methods against both a random baseline and supervised classifiers trained on the same feature representation.

8.2.1 Random Baseline

A random baseline was established by generating 3,000 trials of random binary cluster assignments. This provides a statistical reference point for evaluating whether the clustering algorithms capture meaningful structure in the data.

8.2.2 Supervised Baselines

Two supervised classifiers were trained on the same TFIDF_SVD-5 features used by the best unsupervised configurations:

- **Logistic Regression:** Linear classifier with default parameters (L2 regularization, C=1.0)
- **Support Vector Machine (SVM):** RBF kernel with default parameters

These supervised baselines represent an upper bound on expected performance, as they have direct access to true labels during training.

8.2.3 Results Comparison

Table 6 presents the comprehensive comparison across all methods.

Table 6: Performance Comparison: Unsupervised vs Supervised vs Random Baseline (Test Set)

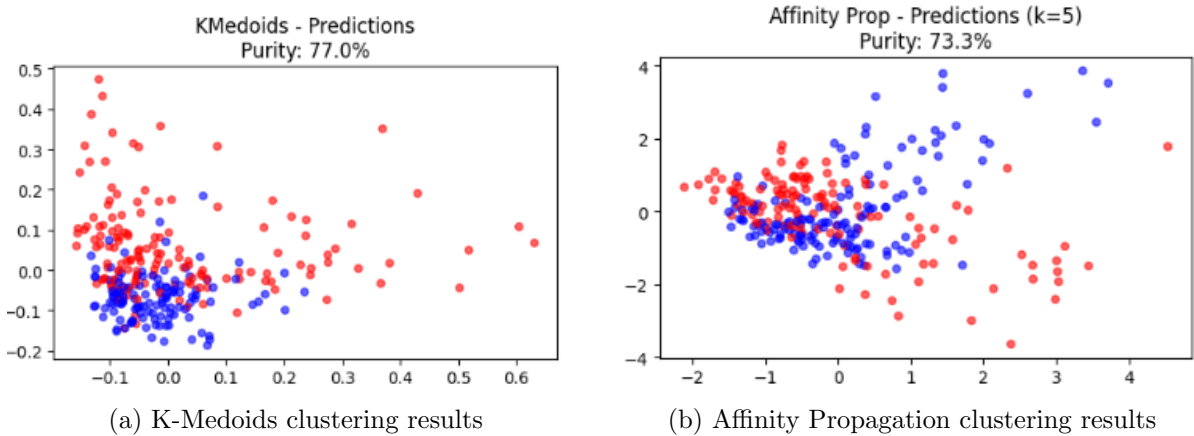
Type	Method	Purity/Acc	ARI	Sil.	Combined
Unsupervised	KMedoids	0.770	0.290	0.157	0.455
	Affinity Propagation	0.733	0.175	0.229	0.409
Supervised	Logistic Regression	0.785	–	–	–
	SVM	0.796	–	–	–
Baseline	Random	0.524	0.000	0.000	0.209

- **Significant improvement over random:** Both unsupervised methods substantially outperformed the random baseline, with KMedoids achieving 77.0% purity compared to 52.4% for random assignment—an improvement of 24.6 percentage points. The combined score more than doubled (0.455 vs 0.209).
- **Competitive with supervised methods:** The unsupervised approaches achieved performance remarkably close to supervised classifiers. KMedoids (77.0% purity) reached 96.7% of the SVM accuracy (79.6%) without using any label information during training.
- **Supervised gap:** The supervised methods outperformed unsupervised clustering by only 2-6 percentage points, suggesting that the TF-IDF features with SVD reduction capture personality-relevant patterns that can be discovered without supervision.

8.3 Cluster Interpretation

Before analyzing the discovered clusters, it is important to acknowledge the inherent difficulty of inferring personality from written text alone. Personality types exhibit substantial overlap, and individuals often adapt their writing style to context, audience, or platform. As a result, linguistic expression may only partially reflect underlying personality traits and can differ from real-life behavior. These factors introduce ambiguity into text-based personality analysis and suggest that even well-separated clusters are expected to contain mixed or transitional samples rather than forming perfectly distinct categories.

To interpret the discovered clusters, we analyze their composition, lexical characteristics, and distinguishing linguistic features, focusing primarily on the K-Medoids solution.



8.3.1 Cluster Composition

Table 7 summarizes the distribution of personality types within each K-Medoids cluster. The results show a clear separation between the two classes: Cluster 0 is predominantly INTJ (79.0%), while Cluster 1 is primarily ENFP (74.9%). Nevertheless, some overlap remains, with misassigned samples reflecting the inherent ambiguity of personality expression in text.

Table 7: Cluster composition for the K-Medoids model (training set).

Cluster	Total	INTJ	ENFP	Majority
0	499	394 (79.0%)	105 (21.0%)	INTJ
1	581	146 (25.1%)	435 (74.9%)	ENFP

8.3.2 Lexical and Linguistic Characteristics

Inspection of the TF-IDF medoids reveals distinct vocabulary usage across clusters. The INTJ-dominant cluster is associated with more measured and analytical language, including hedging terms such as “*somewhat*” and “*maybe*”, as well as references to structured concepts (e.g., “*school*”, “*law*”). In contrast, the ENFP-dominant cluster exhibits a more informal and emotionally expressive style, with frequent contractions (e.g., “*gonna*”), social terms (e.g., “*guys*”, “*dear*”), and affective words (e.g., “*laughing*”, “*warm*”).

Figure 6 further highlights these stylistic differences using handcrafted linguistic features. INTJ-majority texts display higher vocabulary complexity, whereas ENFP-majority texts show increased expressive punctuation usage.

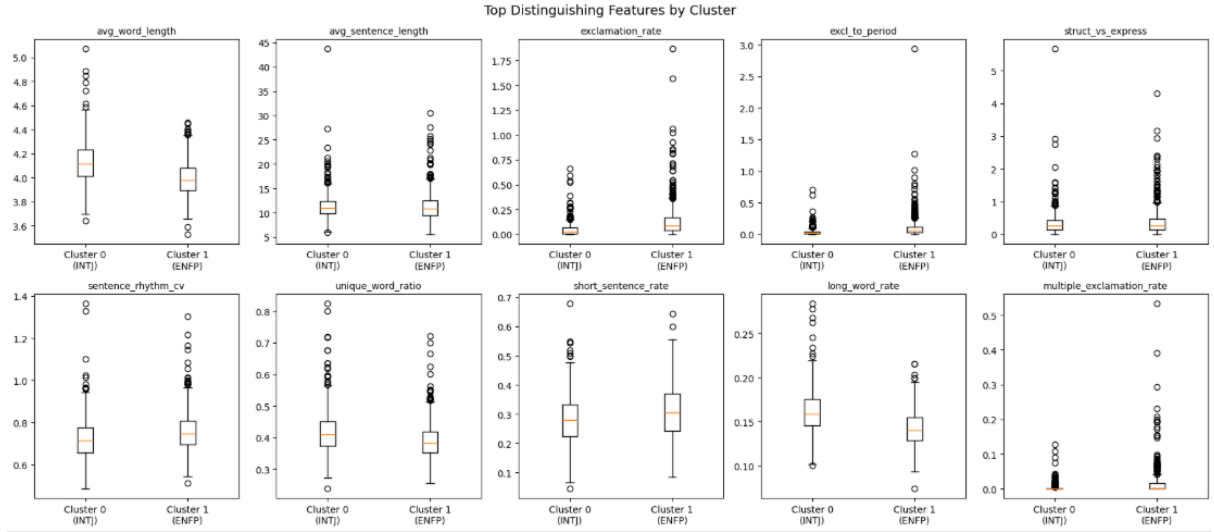


Figure 6: Box plots of the most distinguishing handcrafted linguistic features for each cluster.

8.3.3 Misclassification Analysis

To better understand clustering errors, Table 8 compares misclassified ENFP samples with correctly clustered ENFPs. Misclassified ENFPs use substantially fewer exclamation marks and slightly longer words and sentences, making their writing style closer to that of INTJ users. These cases likely represent natural stylistic variation within personality types.

Table 8: Comparison between misclassified and correctly classified ENFP samples.

Feature	Misclassified	Correct	Difference
exclamation_rate	0.082	0.155	-47.1%
excl.to_period	0.057	0.121	-52.9%
avg_word_length	4.079	3.970	+2.7%
avg_sentence_length	11.626	11.235	+3.5%

8.3.4 Affinity Propagation Analysis

Affinity Propagation identifies five clusters without enforcing a predefined number of groups (Table 9). Two large clusters align closely with the INTJ and ENFP categories and exhibit high purity, while smaller clusters capture more nuanced linguistic subtypes, such as informal expression or engagement with MBTI-related discourse. Although this approach achieves a lower overall purity (73.3%) than K-Medoids (77%), it reveals a richer substructure, suggesting that personality expression in text may lie on a continuum rather than forming strictly binary categories.

Table 9: Cluster composition and purity for Affinity Propagation.

Cluster	Size	INTJ	ENFP	Majority	Purity
0	383	88 (23.0%)	295 (77.0%)	ENFP	77.0%
1	387	306 (79.1%)	81 (20.9%)	INTJ	79.1%
2	68	25 (36.8%)	43 (63.2%)	ENFP	63.2%
3	149	84 (56.4%)	65 (43.6%)	INTJ	56.4%
4	93	35 (37.6%)	58 (62.4%)	ENFP	62.4%

9 Conclusion

This study investigated whether meaningful personality-related structure can be recovered from textual data using unsupervised learning methods. Focusing on the ENFP and INTJ personality types from the MBTI framework, we evaluated two clustering algorithms: K-Medoids and Affinity Propagation across a wide range of feature representations and dimensionality reduction techniques.

Experimental results demonstrate that unsupervised clustering can capture substantial personality signal from text. The best-performing configuration, K-Medoids applied to TF-IDF features reduced via Singular Value Decomposition, achieved a cluster purity of 77% and an ARI of 0.29 on the test set. This performance significantly exceeded a random baseline and approached that of supervised classifiers trained on the same data, which achieved accuracies around 79%.

Among the clustering methods, K-Medoids produced clusters more closely aligned with personality labels, while Affinity Propagation captured finer-grained structure at the cost of reduced purity. Although the results indicate that unsupervised methods can recover meaningful personality-related patterns from text, cluster overlap and misclassifications highlight the inherent ambiguity of inferring personality from writing alone.

Figure 7 summarizes the performance gap between unsupervised and supervised approaches on the test set. While supervised models achieve the highest accuracy, the best unsupervised configuration closely approaches this upper bound without access to label information.

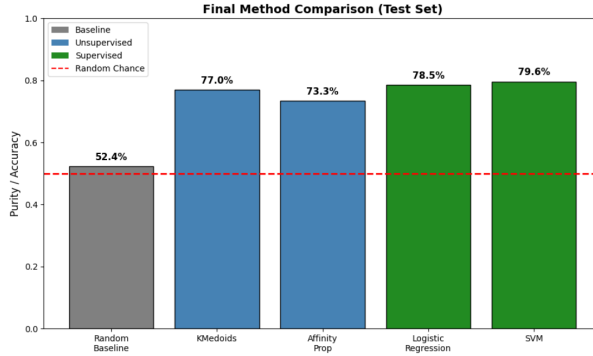


Figure 7: Comparison of supervised classification accuracy and unsupervised clustering purity on the test set.

References

- [1] Brendan J Frey and Delbert Dueck. Clustering by passing messages between data points. *Science*, 315(5814):972–976, 2007.
- [2] Victor Garcia dos Santos and Ivandré Paraboni. Personality traits classification based on text using bert. *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, 2020.
- [3] Hartono, Muhammad Khahfi Zuhanda, Rahmad Syah, Fikriyah Iftinan Fauzi, Nini Sri Wahyuni, Istiana, and Eva Yulina. An adaptive clustering framework for personality prediction using enhanced seed optimization. *Journal of King Saud University – Computer and Information Sciences*, 2022.
- [4] Leonard Kaufman and Peter J Rousseeuw. *Finding groups in data: an introduction to cluster analysis*. John Wiley & Sons, 1990.
- [5] Zeeshan Mushtaq, Ayesha Javed, and Hammad Afzal. Personality trait prediction using machine learning techniques. *IEEE Access*, 9:147160–147173, 2021.
- [6] Kaggle User: Asim Shah. Myers-briggs personality type dataset. <https://www.kaggle.com/datasets/ashishpatel26/myers-briggs-personality-type-dataset>, 2018. Accessed: 2024.